

Monish Prasanna S

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PROFESSIONAL PROFILE

B.Tech. Computer Science student (2027) combining systems-level programming (Rust, C++) with offensive security research. Self-directed experience in: certificate authority workflows, network protocol analysis, and hardware security (ESP32/RFID). Competes in CTFs and builds proof-of-concept tools that bridge **security** and **development**. Looking for internships where secure coding and infrastructure defense are first-class concerns.

EDUCATION

Amrita Vishwa Vidyapeetham

Chennai, India

*B.Tech. in Computer Science and Engineering Specialisation in Cyber Security; **CGPA: 7.34/10***

Expected 2027

Relevant Coursework: Network Security, Cloud Forensics, Cryptography, Compiler Design

TECHNICAL SKILLS

Cybersecurity & Pentesting: Metasploit, Wireshark, websocat, Custom Certificate Authority (CA) Design, Penetration Testing, CAPTCHA Generation Logic, Digital Forensics

Programming & Development: Rust (Memory Safety), Python, C++, PowerShell, Haskell, SQL, Android App Development Basics

Cloud & Infrastructure: Docker, Tailscale, Reverse SSH Tunneling

Networking & IoT: ESP32, RFID/NFC Security, Network Protocol Analysis, Hardware Security

COMPETITIONS & LEADERSHIP

Top 5 Finalist, Finvest — Shastra 2025 | IIT Madras

Jan 2025

- Selected as a Top 5 Finalist in a nationwide pitching competition at India's largest student-run tech fest.
- Developed financial solutions through multi-stage rigorous screening including simulation rounds.

KEY PROJECTS

USB Payload Tool | C, HID, Scripting

- Created a USB-based payload tool that extracts Wi-Fi credentials from Windows hosts and sends captured data to a remote server.
- Automated the complete attack chain from device insertion to credential exfiltration, reducing manual intervention to zero.

Chef Compiler & Security Hub | Compiler Design, Security Documentation

- Designed a 7-phase compiler for the Chef esoteric language and curated a repository of defensive best practices.
- Implemented lexical, syntax, and semantic analysis phases with robust error handling and code generation.

ESP32 RFID Authentication System | Embedded C, IoT Security

- Developed hardware-based auth for audio systems integrating RC522 RFID and I2S hardware security.
- Engineered secure access control with real-time tag verification and unauthorized access logging.

EXPERIENCE & RESEARCH

Independent Cybersecurity Research

Self-Directed

Security Researcher & CTF Participant

Ongoing

- Engineered **Custom Certificate Authority (CA)** workflows to explore PKI security and trust hierarchies.
- Simulated physical-access attacks using **HID emulation** and conducted protocol analysis with **websocat** and **Wireshark**.
- Implemented memory-safe cryptographic protocols (Schnorr signatures) in **Rust**.

Full-Stack URL Shortener Development

Open Source

Lead Developer

Jan 2026 – Mar 2026

- Architected a scalable service using the **MERN stack** and **Redis (Upstash)** for sub-millisecond resolution.
- Implemented rate limiting and security headers to prevent automated abuse and link-scraping.

CERTIFICATIONS

Palo Alto Networks Cybersecurity, Network, Cloud & Security Operations	Feb 2026
Rust Programming Systems Programming Proficiency	Mar 2026